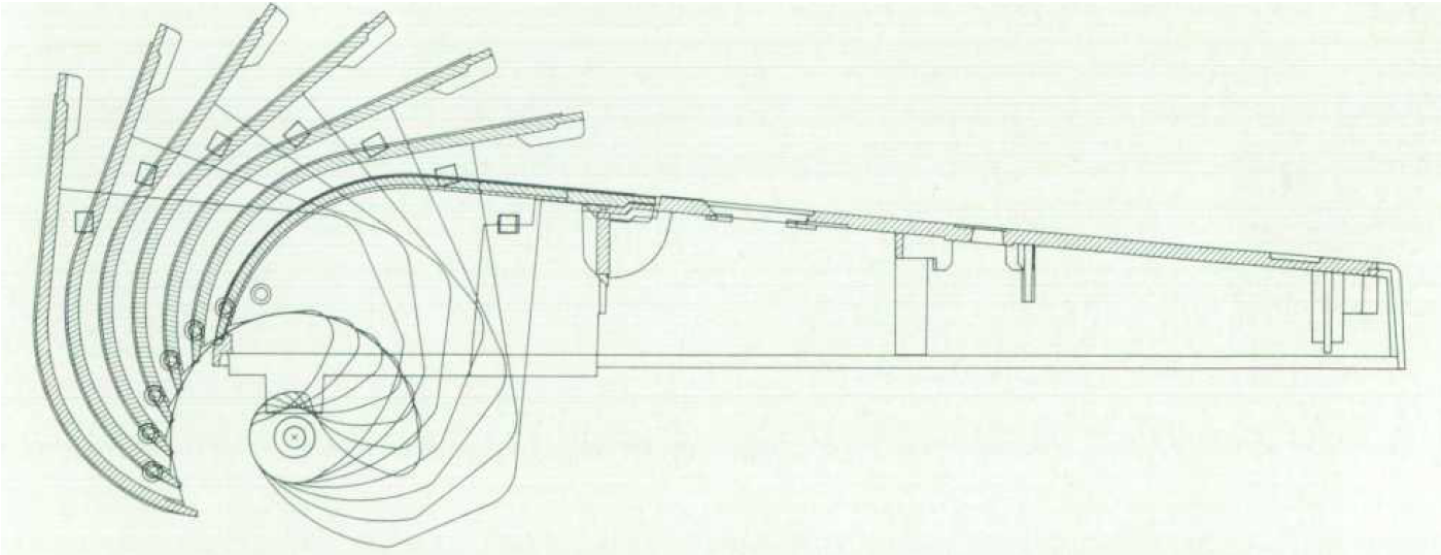


Application note 021: IR Printing



HP-82240A Simulation of paper door rotation used to determine the location of its mounting pin
(HP Journal – October 1987)

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Revision	Date	Description
1.0	2026-04-23	First publication.

Section 1: Overview

Printing to a printer with an Infra Red interface is supported by the C47/R47 through the IR LED provided by the DM hardware. This is an evolution of the HP-42S printing implementation and is focused on remaining compatible with the HP-42S where it makes sense while adding new capabilities to support the new C47/R47 features.

Two kinds of infrared thermal printers are supported:

- HP-82240A/B – This is the original HP IR thermal printer supported by the HP-42S
- Martel 8850/7850 or any other Martel printer supporting the HPIR protocol – These are thermal printers with HP-82240 compatibility and improved speed and resolution. Some industrial printers sold under other brands are manufactured by Martel and may also support the HPIR protocol such as the Midtronics 182-003A or 182-003B.

Note: while the Martel printers listed above are advertised as 82240-compatible, it's not a 100% compatibility as they don't support the 0x04 "PRINT LINE AND LEAVE PRINT HEAD AT RIGHT" control code which is interpreted and printed as "%". This is not an issue for the C47/R47 but it is an issue when printing from the HP-42S for example.

Printing works also in the simulator with a 82240 printer simulator. (See: Appendix C: Printing with the C47 and R47 simulators)

The print annunciator in the status bar comes on whenever the calculator sends information through its infrared interface.

Currently due to ROM space limitations printing is only supported on the DM42n or R47 but not on the original DM42.

Section 2: Menus

The print commands and settings are accessible through the **PRINT** menu.

82240		MAN	NORM	TRACE	STRACE
ON	OFF	#		CHAR	LIST
ADV	STK	x	r	α	PROG

82240		MAN	NORM	TRACE	STRACE
ON	OFF	XFN	ALLr	WIDTH	LCD
ADV	Σ	xy	REGS	TAB	USER

The additional menu allows the selection of the printer being used and includes also commands specific to the printer used. The printer selected is displayed within the menu item.

82240	Martel			MODE	DLAY ₁₈
-------	--------	--	--	------	--------------------

Section 3: Functions list and implementation status

Function	Description	Program -mable	Print to file	Status
82240	Set the printer type to HP-82240	No	-	Done
Martel	Set the printer type to Martel	No	-	Done
MAN	Set Manual mode	Yes	-	Done
NORM	Set Normal mode	Yes	-	Done
TRACE	Set Trace mode	Yes	-	Done
STRACE	Set Trace with stack mode	Yes	-	Done
▣ADV	Advance printer paper	Yes	No	Done
▣ALLr	Print All Registers	Yes	Yes	Done
▣CHAR	Print Character	Yes	No	Done
▣DLAY	Set Delay time between lines	Yes	No	Done
▣LCD	Print LCD Display	Yes	Yes	Done for Martel printer, pending for HP 82240
▣LIST	Print a portion of a program listing	No	No	Done
▣MODE	Set Print Mode	Yes	-	Pending (only mode 0 implemented)
▣OFF	Print Off (clears PRTACT & PR TEN flags)	Yes	-	Done
▣ON	Print On (sets PRTACT & PR TEN flags)	Yes	-	Done
▣PROG	Print Program	No	Yes	Done
▣REGS	Print Register range	Yes	Yes	Done
▣STK	Print Stack	Yes	Yes	Done
▣TAB	Positions the print head to print column n	Yes	No	Done for HP 82240 printer, pending for Martel
▣USER	Print user (variables and programs)	Yes	No	Done
▣WIDTH	Print columns required for an alpha string	Yes	-	Pending
▣α	Print a Register as Alpha	Yes	Yes	Done
▣r	Print a Register	Yes	Yes	Done
▣x	Print X-register	Yes	Yes	Done
▣xy	Print X- and Y-registers as a table	Yes	Yes	Done
▣XFN	Print X, Y and Z registers as a XFN number	Yes	Yes	Pending
▣Σ	Print Summation registers	Yes	No	Done
▣#	Print Byte	Yes	No	Done
	Print annunciator	-	-	Done

When ▣ON is set the printing commands are sent to the infrared interface.

When ▣OFF is set some of the printing commands append data to the file DATA/<YYYYMMDDHHMMSS00>.REGS.TSV in FAT ; a new datafile is generated after a timeout of 2 minutes.

Controlling Printer Operations

**ON**

Printer On

Menu **PRINT** p1 **F1** & p2 **F1**

The **ON** function sets both the **ACT** flag (Printer Active) and **EN** flag (Printer Enable) and enable the infrared printer port.

**OFF**

Printer Off

Menu **PRINT** p1 **F2** & p2 **F2**

The **OFF** function clears both the **ACT** flag (Printer Active) and **EN** flag (Printer Enable) and disable the infrared printer port.

82240

Select HP-82240 as the printer type

Menu **F1**

The **82240** function sets the printer type as HP-82240. Use this for the HP-82240A, HP-82240B or any printer compatible with the HP-82240.

Note: If this setting is used with a Martel HPIR printer there could be some slight variations in the character size or text position depending on the characters printed. This is due to the differences between the characteristics of the Martel printers and the HP-82240A/B.

Martel

Select Martel as the printer type

Menu **F2**

The **Martel** function sets the printer type as Martel. Use this for a Martel printer supporting the HPIR interface. This will ensure that the printing output is optimized for the Martel printers.

**MODE**

Set printer mode

Menu **F5****To be coded.**

 **DLAY *nn*** Set line delay

Menu  **F6**

The  **DLAY** function enables you to specify how long your calculator waits between sending lines of information to the HP 82240A Thermal Printer. The number provided as parameter is the delay time, in ticks. The default DMCP delay is 18, this is the maximum delay value. To optimize printing speed, set the delay time slightly greater than the time it takes for your printer to print a single line of information (this can vary depending on the level of the printer batteries).

 **ADV** Advance

Menu **PRINT** p1  & p2 

When you execute the  **ADV** (advance) function, either in a program or from the keyboard, any information held in the print buffer is printed on the current line, and the printer paper is advanced a single line.

 **TAB *nnn*** Tabulate

Menu **PRINT** p2 

When you execute the  **TAB** (tabulate) function the print head is positioned to print starting at column *nnn*. Note that the number of columns per line depends on the printer type selected : 166 columns for the HP 82240 or 384 columns for a Martel printer.

MAN Set the Manual mode

Menu **PRINT** p1  **F3** & p2  **F3**

In **MAN** mode the printer is idle and does not print unless you or a program executes a print function.

Use this mode when you want the calculator to print only when a print function is executed. The **VIEW** and **AVIEW** functions can also generate printer output in this mode. (This is the default mode.)

NORM Set the Normal mode

Menu **PRINT** p1  **F4** & p2  **F4**

In **NORM** mode the printer prints numbers and ALPHA strings that are keyed in, function names¹ as they are executed from the keyboard, and output from print functions. Running programs print only output from print functions and **PROMPT** output.

Use this mode when you want to print a record of prompts and keystrokes.

¹As with the HP-41C, **ADV** and  **X** function names are not printed.

TRACE

Set the TRACE mode

Menu **PRINT** p1  F5 & p2  F5

In **TRACE** mode the printer prints numbers and ALPHA strings that are keyed in, function names², intermediate and final answers, and output from print functions.

In **TRACE** mode, program listings are printed in a special “packed” or condensed form.

Use this mode when you want to print a record of prompts, keystrokes, and results. If a program is running, each instruction is printed as it executes. This mode is primarily intended for testing and debugging programs.

STRACE

Set the TRACE with stack mode

Menu **PRINT** p1  F6 & p2  F6

The **STRACE** mode is identical to the **TRACE** mode, but instead of the X-register being printed after each operation, the whole stack is printed.

Flags

There are four flags that control printing operations :

- the Printing Mode flags ( **NORM** and  **TRACE**)³
- the Printer Enable flag ( **EN**) and the Printer Active flag ( **ACT**)⁴

The Print Mode Flags determine the print mode of the system:

 TRACE	 NORM	Print Mode
Clear	Clear	Manual
Clear	Set	Normal
Set	Clear	Trace
Set	Set	Trace with stack

The  **ACT** flag can be tested but cannot be set/cleared directly, it is only set or cleared by the functions  **ON** and  **OFF**.

The  **EN** flag is also set by  **ON** and cleared by  **OFF**, and it can be set or cleared by the individual flag commands.

The Printer Enable Flag ( **EN**) is used to control printing in programs that contain specific print functions.  **EN** has no effect on print functions executed from the keyboard. When  **EN** is set, printing functions contained in a program are executed normally. When  **EN** is clear, execution of printing functions in a program is suppressed:

Print commands ( **X**,  **a**,  **r** ...)

 ACT	 EN	Keyboard			Program		
		MAN	NORM	TRACE	MAN	NORM	TRACE
Clear	Clear	No print	No print	No print	Ignored	Ignored	Ignored
Clear	Set	No print	No print	No print	Stop with "Printing Is Disabled"		
Set	Clear	Print	Print	Print	Ignored	Ignored	Ignored
Set	Set	Print	Print	Print	Print	Print	Print

²As with the HP-41C, **ADV** and  **X** function names are not printed.

³ **TRACE** and  **NORM** work as the HP-41C/42s flags 15 and 16 and are set by **MAN**, **NORM** or **TRACE**, or by individual flag commands.

⁴ **EN** and  **ACT** work as the HP-41C/42s flags 21 and 55.

These flags control also how VIEW, AVIEW and PROMPT work when executed from the keyboard or in a program :

VIEW/AVIEW

 ACT	 EN	Keyboard			Program		
		MAN	NORM	TRACE	MAN	NORM	TRACE
Clear	Clear	No print	No print	No print	No print	No print	No print
Clear	Set	No print	No print	No print	STOP	STOP	STOP
Set	Clear	Print	Print	Print	No print	No print	No print
Set	Set	Print	Print	Print	Print	Print	Print

PROMPT

 ACT	 EN	Keyboard			Program		
		MAN	NORM	TRACE	MAN	NORM	TRACE
Clear	Clear	No print	No print	No print	No print	No print	No print
Clear	Set	No print	No print	No print	No print	No print	No print
Set	Clear	No print	Print	Print	No print	No print	No print
Set	Set	No print	Print	Print	No print	Print	Print

Printing Registers

Eight functions enable you to print the contents of different registers or variables. Any listing can be stopped by pressing R/S or EXIT. For the detailed output format related to each data type that can be contained in a register see [Section 3](#).

	Print X	X	data
Menu PRINT p1 			

When you execute the  (print X) function, the content of the X-register is printed and marked with ***. The printed output is right-justified.

Example:

```
3.141592653589E22 ***
```

	Print X and Y tabulated	X	data
		Y	data
Menu PRINT p2 			

When you execute the  (print X and Y) function, the contents of the X-register and of the Y-register are printed on the same line, on the left half for the X-register and on the right half for the Y-register. This is useful to generate tables with pairs of data. If the content of the X-register is a 2xn or nx2 matrix the whole matrix content is printed on two columns.

Limitation: due to the limited printer line width, only 11 characters for each column can be printed , including the number sign and separator(s). Due to this limitation, complex numbers are not supported.

Tip: to print column headers, place the Y column header string in the Y-register, and the X column header string in the X-register and do  before printing the data pairs.

Example:

```

X      Y
1. 4.
2. 5.
3. 6.8

```

	Print a register or variable	Register/variable	data
Menu PRINT p1 			

When you execute the  (print register or variable) function, the content of the register or variable provided as parameter is printed and labeled with the register or variable name. The printed register content is right-justified.

Example:

```
X= 3.141592653590E22
```



Print Alpha

Register/variable

Menu PRINT p1 F5

When you execute the  (print Alpha) function, the content of the register or variable provided as parameter is printed as text. The printed output is left-justified; it lines up with the left margin.

Example:

```
This is a string

compare to the same string printed with r:

X=  "This is a string"
```



Print stack

Menu PRINT p1 F2

D

C

B

A

T

Z

Y

X

When you execute the STK (print stack) function, the contents of the stack registers are printed and labeled in T, Z, Y, X order if SSIZE8 is cleared and in D, C, B, A, T, Z, Y, X order if SSIZE8 is set.

Example:

```
T=      1.254454.5"
Z=  "This is a string"
Y=      A2C8#16
X=      2.718281828459045
```



Print all registers

Menu PRINT p2 GOLD F4

X

Y

⋮

When you execute the ALLr (print all registers) function, the contents of all global registers (A...W, R00...R99) and all currently allocated local registers (if any), are printed and labeled. EXIT or R/S can be pressed at any time to stop the listing.

 REGS	Print registers	R_{ss}	data
			⋮
	$R_{sss+nn-1}$	data	
Menu PRINT p2 F4			

The  **REGS** (print registers) function provides you with control over which registers you print.  **REGS** uses a number in the X register to control printing. Before executing  **REGS**, place a control number in the X-register using the following format: **sss.nn** where **sss** is the starting data storage register address, and **nn** is the number of data storage registers to print. For example, specify registers R10 through R17 by placing 10.08 in the X-register. EXIT or R/S can be pressed at any time to stop the listing.

Example with 10.08 in X:

```

R10="Reg 11,12 & 13 have
: The 3 cubes = 3."
R11=56993682122196238072
0
R12=-5699368211135634935
09
R13= -472715493453327032
R14="Reg 15, 16 & 17 hav
e: The 3 cubes = 42."
R15= -80538738812075974
R16= 80435758145817515
R17= 12602123297335631

```

 Σ	Print statistics registers	n	data
			⋮
	ΣY_{max}	data	
Menu PRINT p2 F2			

The  **Σ** (print statistics registers) function enables you to print the contents of the currently defined statistics registers.

Statistics registers printing example:

```
n= 4.
ΣX= 1824.
ΣY= 492.
ΣX²= 831744.
ΣX²Y= 102304512.
ΣY²= 60516.
ΣXY= 224352.
ΣlnXlnY= 117.8502564553
ΣlnX= 24.4899712380575
Σln²X= 149.939672810221
ΣY·lnX= 3012.2664622811
ΣlnY= 19.2487374214897
Σln²Y= 92.6284730803642
ΣX·lnY= 8777.4242641993
ΣX²lnY= 4002505.4644749
ΣlnY/X= 0.0422121434682
ΣX²/Y= 6762.14634146341
Σ1/X= 0.00877192982456
Σ1/X²= 0.00001923668821
ΣX/Y= 14.8292682926829
Σ1/Y= 0.03252032520325
Σ1/Y²= 0.00026439288783
ΣX³= 379275264.
ΣX⁴= 172949520384.
ΣXmin= 456.
ΣXmax= 456.
ΣYmin= 123.
ΣYmax= 123.
```


Printing Programs

**PROG**

Print program

Menu **PRINT** p1 

When you execute the  **PROG** (print program) function, a specific program stored in program memory is printed. The calculator prompts you for the name of the program you wish to print. Simply key in the name of the program (by pressing [**α**] name [**α**]) or select the program name in the **PROG** menu; printing will begin at the first line of the specified program. If you do not specify a program name in response to the prompt (if you press [**ENTER**]), the printer will print the program to which the calculator is currently positioned, beginning at the first line.

The print mode determines the format in which program lines are printed. You can stop the printing operation at any time by pressing [**EXIT**] or [**R/S**]. This function is not programmable.

With TRACE mode disabled each program instruction is printed left-justified on a separate line starting with the program step number.

With TRACE mode enabled the program is printed in a condensed form with each label instruction on a separate line preceded by a blank line and with the following instructions just separated by two spaces without step numbers. (This format was supported by the HP 41 but not by the HP 42S).

The time and date are also printed at the head of the program listing (as for the HP-41CX or 41C/CV with the time module).

Sample program printed in MAN or NORM mode:

```
07:58 2025-11-11
00 ( 72-Byte Prgm )
01▶LBL 'M.Diag'
02 LOCR 08
03 STOS .00
04 M.DIM?
05 x# ? Y
06 GTO 00
07 INDEX .00
08 STOIJ
09 1
10▶LBL 01
11 RCLEL
12 Y<> I
13 STOEL
14 Y<> I
15 DROPx
16 I-
17 J-
18 DSE Y
19 GTO 01
20 RCL Z
21 M.DIM .00
22 RCLS .00
23 RTN
24▶LBL 00
25 RCLS .00
26 ERR 21
27 END
```

Same program printed in TRACE mode:

```
07:57 2025-11-11
00 ( 72-Byte Prgm )
01▶LBL 'M.Diag'
LOC 08 STOS .00
M.DIM? x# ? Y GTO 00
INDEX .00 STOIJ 1

10▶LBL 01
RCLEL Y<> I STOEL
Y<> I DROPx I- J-
DSE Y GTO 01 RCL Z
M.DIM .00 RCLS .00 RTN

24▶LBL 00
RCLS .00 ERR 21 END
```

 **LIST** *nnnn* Print part of a program

Menu **PRINT** p1  

When you execute the  **LIST** (print listing) function, the specified number of lines are printed starting from the current program pointer.

First, position the calculator to the desired program and then to the line within the program where you wish printing to begin. Then execute  **LIST**.

The program listing begins with the current program line and continues for *nnnn* lines or until an END instruction is encountered.

You can stop the printing operation at any time by pressing [EXIT] or [R/S]. This function is not programmable.

 **LIST 0008** from step 07 in MAN or NORM mode:

```
07 INDEX .00
08 STOIJ
09 1
10 LBL 01
11 RCLEL
12 Y<> I
13 STOEL
14 Y<> I
```

 **LIST 0008** from step 07 in TRACE mode:

```
ALIST 0008
INDEX .00 STOIJ 1
10 LBL 01
RCLEL Y<> I STOEL
Y<> I
```

Printing Characters

Two functions enable you to send single characters to the printer.

 **#** *nnn* Print byte

Menu **PRINT** p1  

The  **#** (print byte) function sends to the printer a single byte (value between 0 and 255) provided as the function parameter. This value is either mapped by the printer to its character set or can be used for specific control command such as ESC sequences. Refer to the printer reference manual for specific details.

 **CHAR** Print character

Register/variable

Menu **PRINT** p1  

The  **CHAR** (print character) function sends to the printer a single character whose Unicode value is in the register specified. Only characters that are in the C47/R47 standard font are supported. This Unicode value is either mapped to the printer character set if possible (see [Appendix B](#)), or printed as graphic within the printer resolution capabilities.

Other Printing Commands



Print LCD display

Menu **PRINT** p2

The (print LCD) function prints a full copy of the display on a Martel printer and only a subset of the display (the stack area) on the HP 82240 printer due to its lower resolution. This is a full graphic print and as such is pretty slow.



Print User variables and program labels

Menu **PRINT** p2

The (print user) function prints all user variable names and values, followed by the list of global program labels. The variables are printed first. EXIT or R/S can be pressed at any time to stop the listing.

Reserved variables are not printed by . If needed they can be printed individually with .

User variables are printed following their order in memory, same for the program labels. In addition to the global labels, the END instructions are also printed, which helps to identify global labels that are in the same program, as well as programs without a global label. The program number is printed with the first global label of each program, if there is no global label, the program number is printed with the END instruction.

Example:

```
aa=          13
bb=          456
cc=          789

LBL 'ATEST'  Prgm #1/4
END
LBL 'ABC'    Prgm #2/4
LBL 'GHI'
LBL 'EAE'
END
END          Prgm #3/4
LBL 'M.Diag' Prgm #4/4
.END.
```



Number of columns used to print a string

Register/variable

Menu **PRINT** p2

The (print width) function returns the number of columns that will be used to print a string in a register or variable provided as parameter. This number depends on the printer type selected (HP 82240 or Martel) - and possibly of the print mode.

Section 4: Printing output for the different data types

The C47/R47 supports more data types than the HP calculators for which the 82240 thermal printer was designed. Here is how the printing output of the different data types is adjusted to the limitations of the 82240 printer.

Long Integer

Long integers are fully printed and can overlap on several lines depending of the number of digits.

Examples:

```
1234567890123456   ***
123456789012345678912345
689123456789       ***
```

Real number

Real numbers are printed in a format that ensure they fit in a single line. The exponent is indicated by the letter E. The different angle modes are also printed in a format similar to the display one, with some adjustments to fit the printer resolution. *(Currently the various separators are not supported)*

Examples:

```
-1.234567891E-456   ***
X= -1.234567891E-456
```

output for degree, radian, grad, DMS and multPi :

```
180.45°   ***
3.149446635223768°   ***
200.5°   ***
180°27' 0.00"   ***
1.0025"   ***
```

Complex number

For complex numbers each part (real and imaginary) is printed as a real number, so they may overlap on two lines.

Examples:

```
12.345+i×4.56E91   ***
12.345∠11.34°   ***
```

Time

Times are printed as they appear on the display in the *hours:minutes:seconds.decimals* format.

Examples:

```
14:59:27   ***
14:59:36.648   ***
```

Date

Dates are printed as they appear on the display in the DMY, MDY or YMD format.

Examples:

```
11/08/2025 ***
08.11.2025 ***
2025-11-08 ***
```

String

Strings are printed in two different ways: as simple text left justified when printed with α , or within double quotes and right justified when printed with any other print command. The characters from the C47/R47 standard font are mapped to the printer font if available, otherwise they are printed as graphic but due to the limited resolution of the printer there is a loss of quality compared to the rendering on the display.

Examples:

'This is a string' in X, printed with α X:

```
This is a string
```

'This is a string' in X, printed with α r X:

```
X= "This is a string"
```

Real matrix

A real matrix is printed with the format used by the HP 42S, with a header giving the matrix dimension followed by the list of the elements line by line.

Examples:

```
R00= [ 2x2 Matrix ]
1:1= 1.23456789E-456
1:2= 2.31645978E456
2:1= 3.1234568E-4562
2:2= 4567892356000.
```

Complex matrix

A complex matrix is printed with a format similar to a real matrix, but the header indicates that it's a complex matrix and each element is printed as a complex number and may overlap on two lines.

Examples:

```
R01= [ 2x2 Cpx Matrix ]
1:1=1.23456789E-456+ix1.
76009744768817
1:2=2.31645978E456+ix3.3
02527937E912
2:1=3.1234568E-4562+ix4.
4530466E-4106
2:2=4567892356000.+ix6.5
12347958E468
```

```
X= [ 2x2 Cpx Matrix ]
1:1= 1.2445.9
1:2= 2.345.59
2:1= 0.4564102.9
2:2= 5.42489.9
```

Short Integer

Short integers are printed as long integers but with *#base* added to indicate the base used. They can overlap on several lines if they don't fit on one.

Examples:

```
123456ABCDEF#16 ***
123456ABCDEF4321#16
***
110642552571573641441#8
***
100100011010001010110101
010111100110111101111010
0001100100001#2 ***
```

Configuration data

When printing a register containing configuration data (saved with STOCFG), the register content which is not printable is replaced by the text 'Config. Data'.

Example:

```
R09= Config. data
```

Appendix A: HP-42s and C47/R47 Print commands and flags comparison

	HP-42S	Description	C47/R47	Description
Commands	ADV	Advances the printer paper one line.	ADV	Prints the current contents of the print buffer plus an LF.
	-		CHAR	Sends a single character (with the Unicode specified) to the printer.
	DELAY	Sets the print <i>delay</i> time to x seconds.	DELAY	Sets a delay of n ticks (see TICKS) to be used with each linefeed on the printer.
	LIST	Prints a portion of a program listing. (Not programmable.) Parameter: number of lines Indirect: No	LIST	Prints a portion of a program listing.
	-		MODE	Sets printer mode. (Details TBD)
	MAN	Selects <i>Manual</i> print mode.	MAN	Selects <i>Manual</i> print mode.
	NORM	Selects <i>Normal</i> print mode, which prints a record of keystrokes.	NORM	Selects <i>Normal</i> print mode, which prints a record of keystrokes.
	PRA	Print <i>Alpha register</i> as text, left justified.	α	Appends alpha to the print line, trailed by an LF.
	PRLCD	Print <i>LCD (liquid crystal display)</i> . Prints the entire display.	LCD	Full LCD display printing supported on a Martel printer, only the stack registers display area printing is supported on the 82240 due to the lower printer resolution.
	PROFF	Printing off. Clears flags 21 and 55.	OFF	Printing Off. Clears flags PR TEN and PR TACT.
	PRON	Printing on. Sets flags 21 and 55.	ON	Printing On. Sets flags PR TEN and PR TACT.
	PRP	Print <i>program</i> . If a label is not specified, prints the current program. (Not programmable.) Parameter: global label (optional) Indirect: No	PROG	Prints the listing of the specified or current program, one line per step.
	PRSTK	Print <i>stack</i> . Prints the contents of the stack registers (X, Y, Z, and T).	STK	Prints the entire stack contents. Each of the 4 or 8 registers prints in a separate row starting with its label.
	PRUSR	Prints user variables and programs.	USER	Prints all variables and global program labels in alphabetic order.
	PRV	Print <i>variable</i> .	r	Prints the content of the Register or Variable specified, right adjusted with a heading label.
	PRX	Print <i>X-register</i> .	x	Prints x without a heading label.
	-		xy	Prints x and y without a heading label.
	PRΣ	Print <i>statistics</i> . Prints the contents of the summation registers.	Σ	Prints the summation registers. Each register prints in one row starting with its label.
	-		REGS	Interprets x in the form sss.nn . Prints the contents of nn registers starting with number sss . Each register takes one row starting with its label.
	-		TAB	Positions the print head to print column n (0 to 165). Useful in formatting. Allows also for printer plots. If n is less than current print head position, a linefeed will be entered to reach the new position.
	TRACE	Selects <i>Trace</i> printing mode, which prints a record of keystrokes and results.	TRACE	Selects <i>Trace</i> printing mode, which prints a record of keystrokes and results.
	-		WIDTH	Returns the number of print columns that alpha would take in the print mode set.
	-		#	Sends a single byte, without translation, to the printer (e.g., a control code).

Flags	Flag 12	<i>Double-Wide Printer Output.</i> If flag 12 is set, all printer output is printed double-wide.		-	<i>Is it needed ??</i>
	Flag 13	<i>Lowercase Printer Output.</i> If flag 13 is set, the letters A through Z are printed in lowercase.		-	<i>This was needed on the 42s for compatibility with the HP-41 which didn't have a full lowercase character set. Dropped on the C47/R47.</i>
	Flags 15 & 16	<i>Print Mode.</i> This table shows how flags 15 and 16 represent the current print mode.		TRACE NORM	<i>Print Mode.</i> This table shows how flags TRACE and NORM represent the current print mode.
	Flag 21	<i>Printer Enable.</i> Flag 21 allows your program to control how the functions VIEW and AVIEW are executed.		EN	<i>Printer Enable.</i> Flag EN allows your program to control how the functions VIEW and AVIEW are executed. This flag determines if printing is <i>desired</i> .
	Flag 55	<i>Printer Existence.</i> Executing the PRON (printing on) function enables printing by setting flags 21 and 55. Executing PROFF (printing off) disables printing and clears flags 21 and 55.		ACT	<i>Printer Active.</i> Flag ACT is a system flag, it cannot be directly set or cleared. It can be tested. Executing the ON (printing on) function enables printing by setting flags EN and ACT. Executing OFF (printing off) disables printing and clears flags EN and ACT. This flag determines if printing is <i>possible</i> .

Appendix B: HP 82240 ROMAN-8 Characters Mapping To Unicode

An asterisk (*) indicates that the corresponding character is not available in the C47/R47 font.

0x00,	0x0000,	#NULL	0x40,	0x0040,	#COMMERCIAL AT
0x01,	0x0000,	#UNDEFINED	0x41,	0x0041,	#LATIN CAPITAL LETTER A
0x02,	0x0000,	#UNDEFINED	0x42,	0x0042,	#LATIN CAPITAL LETTER B
0x03,	0x0000,	#UNDEFINED	0x43,	0x0043,	#LATIN CAPITAL LETTER C
0x04,	0x0004,	#PRINT LINE AND LEAVE PRINT HEAD AT RIGHT	0x44,	0x0044,	#LATIN CAPITAL LETTER D
0x05,	0x0000,	#UNDEFINED	0x45,	0x0045,	#LATIN CAPITAL LETTER E
0x06,	0x0000,	#UNDEFINED	0x46,	0x0046,	#LATIN CAPITAL LETTER F
0x07,	0x0000,	#UNDEFINED	0x47,	0x0047,	#LATIN CAPITAL LETTER G
0x08,	0x0000,	#UNDEFINED	0x48,	0x0048,	#LATIN CAPITAL LETTER H
0x09,	0x0000,	#UNDEFINED	0x49,	0x0049,	#LATIN CAPITAL LETTER I
0x0A,	0x000A,	#LINE FEED	0x4A,	0x004A,	#LATIN CAPITAL LETTER J
0x0B,	0x0000,	#UNDEFINED	0x4B,	0x004B,	#LATIN CAPITAL LETTER K
0x0C,	0x0000,	#UNDEFINED	0x4C,	0x004C,	#LATIN CAPITAL LETTER L
0x0D,	0x0000,	#UNDEFINED	0x4D,	0x004D,	#LATIN CAPITAL LETTER M
0x0E,	0x0000,	#UNDEFINED	0x4E,	0x004E,	#LATIN CAPITAL LETTER N
0x0F,	0x0000,	#UNDEFINED	0x4F,	0x004F,	#LATIN CAPITAL LETTER O
0x10,	0x0000,	#UNDEFINED	0x50,	0x0050,	#LATIN CAPITAL LETTER P
0x11,	0x0000,	#UNDEFINED	0x51,	0x0051,	#LATIN CAPITAL LETTER Q
0x12,	0x0000,	#UNDEFINED	0x52,	0x0052,	#LATIN CAPITAL LETTER R
0x13,	0x0000,	#UNDEFINED	0x53,	0x0053,	#LATIN CAPITAL LETTER S
0x14,	0x0000,	#UNDEFINED	0x54,	0x0054,	#LATIN CAPITAL LETTER T
0x15,	0x0000,	#UNDEFINED	0x55,	0x0055,	#LATIN CAPITAL LETTER U
0x16,	0x0000,	#UNDEFINED	0x56,	0x0056,	#LATIN CAPITAL LETTER V
0x17,	0x0000,	#UNDEFINED	0x57,	0x0057,	#LATIN CAPITAL LETTER W
0x18,	0x0000,	#UNDEFINED	0x58,	0x0058,	#LATIN CAPITAL LETTER X
0x19,	0x0000,	#UNDEFINED	0x59,	0x0059,	#LATIN CAPITAL LETTER Y
0x1A,	0x0000,	#UNDEFINED	0x5A,	0x005A,	#LATIN CAPITAL LETTER Z
0x1B,	0x001B,	#ESCAPE	0x5B,	0x005B,	#LEFT SQUARE BRACKET
0x1C,	0x0000,	#UNDEFINED	0x5C,	0x005C,	#REVERSE SOLIDUS
0x1D,	0x0000,	#UNDEFINED	0x5D,	0x005D,	#RIGHT SQUARE BRACKET
0x1E,	0x0000,	#UNDEFINED	0x5E,	0x005E,	#CIRCUMFLEX ACCENT
0x1F,	0x0000,	#UNDEFINED	0x5F,	0x005F,	#LOW LINE
0x20,	0x0020,	#SPACE	0x60,	0x2019,	#RIGHT SINGLE QUOTATION MARK *
0x21,	0x0021,	#EXCLAMATION MARK	0x61,	0x0061,	#LATIN SMALL LETTER A
0x22,	0x0022,	#QUOTATION MARK	0x62,	0x0062,	#LATIN SMALL LETTER B
0x23,	0x0023,	#NUMBER SIGN	0x63,	0x0063,	#LATIN SMALL LETTER C
0x24,	0x0024,	#DOLLAR SIGN	0x64,	0x0064,	#LATIN SMALL LETTER D
0x25,	0x0025,	#PERCENT SIGN	0x65,	0x0065,	#LATIN SMALL LETTER E
0x26,	0x0026,	#AMPERSAND	0x66,	0x0066,	#LATIN SMALL LETTER F
0x27,	0x0027,	#APOSTROPHE	0x67,	0x0067,	#LATIN SMALL LETTER G
0x28,	0x0028,	#LEFT PARENTHESIS	0x68,	0x0068,	#LATIN SMALL LETTER H
0x29,	0x0029,	#RIGHT PARENTHESIS	0x69,	0x0069,	#LATIN SMALL LETTER I
0x2A,	0x002A,	#ASTERISK	0x6A,	0x006A,	#LATIN SMALL LETTER J
0x2B,	0x002B,	#PLUS SIGN	0x6B,	0x006B,	#LATIN SMALL LETTER K
0x2C,	0x002C,	#COMMA	0x6C,	0x006C,	#LATIN SMALL LETTER L
0x2D,	0x002D,	#HYPHEN-MINUS	0x6D,	0x006D,	#LATIN SMALL LETTER M
0x2E,	0x002E,	#FULL STOP	0x6E,	0x006E,	#LATIN SMALL LETTER N
0x2F,	0x002F,	#SOLIDUS	0x6F,	0x006F,	#LATIN SMALL LETTER O
0x30,	0x0030,	#DIGIT ZERO	0x70,	0x0070,	#LATIN SMALL LETTER P
0x31,	0x0031,	#DIGIT ONE	0x71,	0x0071,	#LATIN SMALL LETTER Q
0x32,	0x0032,	#DIGIT TWO	0x72,	0x0072,	#LATIN SMALL LETTER R
0x33,	0x0033,	#DIGIT THREE	0x73,	0x0073,	#LATIN SMALL LETTER S
0x34,	0x0034,	#DIGIT FOUR	0x74,	0x0074,	#LATIN SMALL LETTER T
0x35,	0x0035,	#DIGIT FIVE	0x75,	0x0075,	#LATIN SMALL LETTER U
0x36,	0x0036,	#DIGIT SIX	0x76,	0x0076,	#LATIN SMALL LETTER V
0x37,	0x0037,	#DIGIT SEVEN	0x77,	0x0077,	#LATIN SMALL LETTER W
0x38,	0x0038,	#DIGIT EIGHT	0x78,	0x0078,	#LATIN SMALL LETTER X
0x39,	0x0039,	#DIGIT NINE	0x79,	0x0079,	#LATIN SMALL LETTER Y
0x3A,	0x003A,	#COLON	0x7A,	0x007A,	#LATIN SMALL LETTER Z
0x3B,	0x003B,	#SEMICOLON	0x7B,	0x007B,	#LEFT CURLY BRACKET
0x3C,	0x003C,	#LESS-THAN SIGN	0x7C,	0x007C,	#VERTICAL LINE
0x3D,	0x003D,	#EQUALS SIGN	0x7D,	0x007D,	#RIGHT CURLY BRACKET
0x3E,	0x003E,	#GREATER-THAN SIGN	0x7E,	0x007E,	#TILDE
0x3F,	0x003F,	#QUESTION MARK	0x7F,	0x0000,	#UNDEFINED *

0x80, 0X00A0, #NO-BREAK SPACE	0xC0, 0x00E2, #LATIN SMALL LETTER A WITH CIRCUMFLEX
0x81, 0x00F7, #DIVISION SIGN	0xC1, 0x00EA, #LATIN SMALL LETTER E WITH CIRCUMFLEX
0x82, 0x00D7, #MULTIPLICATION SIGN	0xC2, 0x00F4, #LATIN SMALL LETTER O WITH CIRCUMFLEX
0x83, 0x221A, #SQUARE ROOT	0xC3, 0x00FB, #LATIN SMALL LETTER U WITH CIRCUMFLEX
0x84, 0x222B, #INTEGRAL	0xC4, 0x00E1, #LATIN SMALL LETTER A WITH ACUTE
0x85, 0x2211, #N-ARY SUMMATION	0xC5, 0x00E9, #LATIN SMALL LETTER E WITH ACUTE
0x86, 0x25B6, #BLACK RIGHT-POINTING TRIANGLE	* 0xC6, 0x00F3, #LATIN SMALL LETTER O WITH ACUTE
0x87, 0x03C0, #GREEK SMALL LETTER PI	0xC7, 0x00FA, #LATIN SMALL LETTER U WITH ACUTE
0x88, 0x2202, #PARTIAL DIFFERENTIAL	0xC8, 0x00E0, #LATIN SMALL LETTER A WITH GRAVE
0x89, 0x2264, #LESS-THAN OR EQUAL TO	0xC9, 0x00E8, #LATIN SMALL LETTER E WITH GRAVE
0x8A, 0x2265, #GREATER-THAN OR EQUAL TO	0xCA, 0x00F2, #LATIN SMALL LETTER O WITH GRAVE
0x8B, 0x2260, #NOT EQUAL TO	0xCB, 0x00F9, #LATIN SMALL LETTER U WITH GRAVE
0x8C, 0x03B1, #GREEK SMALL LETTER ALPHA	0xCC, 0x00E4, #LATIN SMALL LETTER A WITH DIAERESIS
0x8D, 0x2192, #RIGHTWARDS ARROW	0xCD, 0x00EB, #LATIN SMALL LETTER E WITH DIAERESIS
0x8E, 0x2190, #LEFTWARDS ARROW	0xCE, 0x00F6, #LATIN SMALL LETTER O WITH DIAERESIS
0x8F, 0x03BC, #GREEK SMALL LETTER MU	0xCF, 0x00FC, #LATIN SMALL LETTER U WITH DIAERESIS
0x90, 0x21B5, #DOWNWARDS ARROW WITH CORNER LEFTWARDS	0xD0, 0x00C5, #LATIN CAPITAL LETTER A WITH RING ABOVE
0x91, 0x00B0, #DEGREE SIGN	0xD1, 0x00EE, #LATIN SMALL LETTER I WITH CIRCUMFLEX
0x92, 0x00AB, #LEFT-POINTING DOUBLE ANGLE QUOTATION MARK	0xD2, 0x00D8, #LATIN CAPITAL LETTER O WITH STROKE
0x93, 0x00BB, #RIGHT-POINTING DOUBLE ANGLE QUOTATION MARK	0xD3, 0x00C6, #LATIN CAPITAL LETTER AE
0x94, 0x22A2, #RIGHT TACK	0xD4, 0x00E5, #LATIN SMALL LETTER A WITH RING ABOVE
0x95, 0x2081, #SUBSCRIPT ONE	0xD5, 0x00ED, #LATIN SMALL LETTER I WITH ACUTE
0x96, 0x2082, #SUBSCRIPT TWO	0xD6, 0x00F8, #LATIN SMALL LETTER O WITH STROKE
0x97, 0x2162, #SUPERSCRIPIT TWO (REDEFINED)	0xD7, 0x00E6, #LATIN SMALL LETTER AE
0x98, 0x2163, #SUPERSCRIPIT THREE (REDEFINED)	0xD8, 0x00C4, #LATIN CAPITAL LETTER A WITH DIAERESIS
0x99, 0x24A4, #SUBSCRIPT LATIN SMALL LETTER I (REDEFINED)	0xD9, 0x00EC, #LATIN SMALL LETTER I WITH GRAVE
0x9A, 0x24A5, #SUBSCRIPT LATIN SMALL LETTER J (REDEFINED)	0xDA, 0x00D6, #LATIN CAPITAL LETTER O WITH DIAERESIS
0x9B, 0x0000, #UNDEFINED	* 0xDB, 0x00DC, #LATIN CAPITAL LETTER U WITH DIAERESIS
0x9C, 0x248A, #SUPERSCRIPIT LATIN SMALL LETTER I (REDEFINED)	0xDC, 0x00C9, #LATIN CAPITAL LETTER E WITH ACUTE
0x9D, 0x248B, #SUPERSCRIPIT LATIN SMALL LETTER J (REDEFINED)	0xDD, 0x00EF, #LATIN SMALL LETTER I WITH DIAERESIS
0x9E, 0x248C, #SUPERSCRIPIT LATIN SMALL LETTER K (REDEFINED)	0xDE, 0x00DF, #LATIN SMALL LETTER SHARP S (GERMAN)
0x9F, 0x248F, #SUPERSCRIPIT LATIN SMALL LETTER N (REDEFINED)	0xDF, 0x00D4, #LATIN CAPITAL LETTER O WITH CIRCUMFLEX
0xA0, 0x2221, #MEASURED ANGLE	0xE0, 0x00C1, #LATIN CAPITAL LETTER A WITH ACUTE
0xA1, 0x00C0, #LATIN CAPITAL LETTER A WITH GRAVE	0xE1, 0x00C3, #LATIN CAPITAL LETTER A WITH TILDE
0xA2, 0x00C2, #LATIN CAPITAL LETTER A WITH CIRCUMFLEX	0xE2, 0x00E3, #LATIN SMALL LETTER A WITH TILDE
0xA3, 0x00C8, #LATIN CAPITAL LETTER E WITH GRAVE	0xE3, 0x00D0, #LATIN CAPITAL LETTER ETH (ICELANDIC)
0xA4, 0x00CA, #LATIN CAPITAL LETTER E WITH CIRCUMFLEX	* 0xE4, 0x00F0, #LATIN SMALL LETTER ETH (ICELANDIC)
0xA5, 0x00CB, #LATIN CAPITAL LETTER E WITH DIAERESIS	* 0xE5, 0x00CD, #LATIN CAPITAL LETTER I WITH ACUTE
0xA6, 0x00CE, #LATIN CAPITAL LETTER I WITH CIRCUMFLEX	0xE6, 0x00CC, #LATIN CAPITAL LETTER I WITH GRAVE
0xA7, 0x00CF, #LATIN CAPITAL LETTER I WITH DIAERESIS	0xE7, 0x00D3, #LATIN CAPITAL LETTER O WITH ACUTE
0xA8, 0x00B4, #ACUTE ACCENT	* 0xE8, 0x00D2, #LATIN CAPITAL LETTER O WITH GRAVE
0xA9, 0x0060, #GRAVE ACCENT	* 0xE9, 0x00D5, #LATIN CAPITAL LETTER O WITH TILDE
0xAA, 0x005E, #CIRCUMFLEX ACCENT	* 0xEA, 0x00F5, #LATIN SMALL LETTER O WITH TILDE
0xAB, 0x00A8, #DIAERESIS	* 0xEB, 0x0160, #LATIN CAPITAL LETTER S WITH CARON
0xAC, 0x02DC, #SMALL TILDE	* 0xEC, 0x0161, #LATIN SMALL LETTER S WITH CARON
0xAD, 0x00D9, #LATIN CAPITAL LETTER U WITH GRAVE	0xED, 0x00DA, #LATIN CAPITAL LETTER U WITH ACUTE
0xAE, 0x00FB, #LATIN SMALL LETTER U WITH CIRCUMFLEX	0xEE, 0x0178, #LATIN CAPITAL LETTER Y WITH DIAERESIS
0xAF, 0X20A4, #LIRA SIGN	* 0xEF, 0x00FF, #LATIN SMALL LETTER Y WITH DIAERESIS
0xB0, 0x00AF, #MACRON	* 0xF0, 0x00DE, #LATIN CAPITAL LETTER THORN (ICELANDIC)
0xB1, 0x00DD, #LATIN CAPITAL LETTER Y WITH ACUTE	* 0xF1, 0x00FE, #LATIN SMALL LETTER THORN (ICELANDIC)
0xB2, 0x00FD, #LATIN SMALL LETTER Y WITH ACUTE	* 0xF2, 0x00B7, #MIDDLE DOT
0xB3, 0x00B0, #DEGREE SIGN	0xF3, 0x00B5, #MICRO SIGN
0xB4, 0x00C7, #LATIN CAPITAL LETTER C WITH CEDILLA	0xF4, 0x00B6, #PILCROW SIGN
0xB5, 0x00E7, #LATIN SMALL LETTER C WITH CEDILLA	0xF5, 0x00BE, #VULGAR FRACTION THREE QUARTERS
0xB6, 0x00D1, #LATIN CAPITAL LETTER N WITH TILDE	0xF6, 0x00AD, #SOFT HYPHEN
0xB7, 0x00F1, #LATIN SMALL LETTER N WITH TILDE	0xF7, 0x00BC, #VULGAR FRACTION ONE QUARTER
0xB8, 0x00A1, #INVERTED EXCLAMATION MARK	* 0xF8, 0x00BD, #VULGAR FRACTION ONE HALF
0xB9, 0x00BF, #INVERTED QUESTION MARK	0xF9, 0x00AA, #FEMININE ORDINAL INDICATOR
0xBA, 0x00A4, #CURRENCY SIGN	* 0xFA, 0x00BA, #MASCULINE ORDINAL INDICATOR
0xBB, 0x00A3, #POUND SIGN	* 0xFB, 0x00AB, #LEFT-POINTING DOUBLE ANGLE QUOTATION MARK
0xBC, 0x00A5, #YEN SIGN	0xFC, 0x25AE, #BLACK VERTICAL RECTANGLE
0xBD, 0x00A7, #SECTION SIGN	0xFD, 0x00BB, #RIGHT-POINTING DOUBLE ANGLE QUOTATION MARK
0xBE, 0x0192, #LATIN SMALL LETTER F WITH HOOK	* 0xFE, 0x00B1, #PLUS-MINUS SIGN
0xBF, 0x00A2, #CENT SIGN	0xFF, 0x0000, #UNDEFINE

Appendix C: Printing with the C47 and R47 simulators

The printing functions are supported by the C47 and R47 simulators, the print data is transmitted over UDP (the native interface of the Emu series emulators). This makes the C47 and R47 printing functions compatible with existing HP-82240 simulators using this interface.

Two printer simulators have been confirmed working with the C47 and R47 simulators:

- The HP82240B Printer Simulator from Christoph Giesselink : <https://hp.giesselink.com/hp82240b.htm>

This simulator provides two windows, one displaying only the text data sent to the printer, and a second graphic window displaying text and graphic characters as on the physical HP 82240B printer. This second window is the one providing the full printing output as many characters from the C47 font are sent as graphics to the printer.

- The Qt Printer Emulator developed for the WP 34s project by Pascal Meheut and included in the latest releases for Windows, MacOS or Linux : <https://sourceforge.net/projects/wp34s/files/emulator/>

This Qt Emulator has been tested on Windows and Linux and works correctly with the C47 and R47 simulators, however the default size of the print window is quite small compared to the display of the calculator simulator so it needs to be adjusted to get bigger characters.

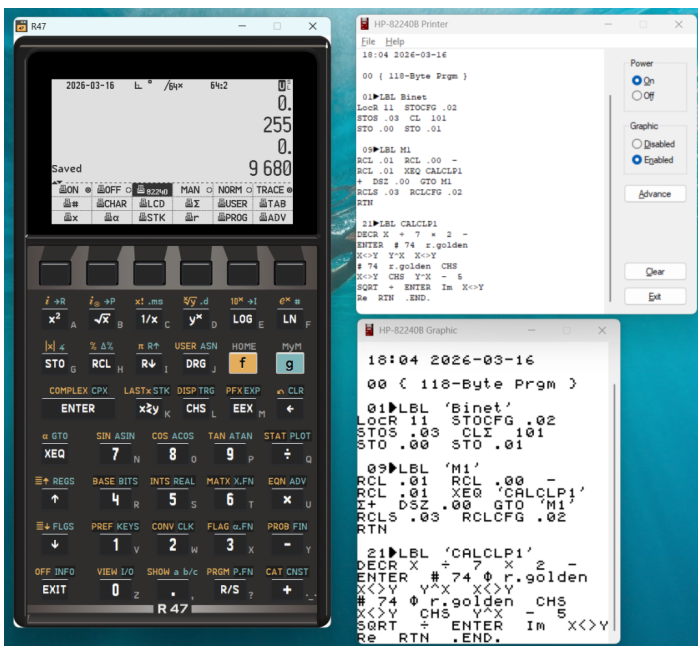


Figure 1: HP82240B Printer Simulator

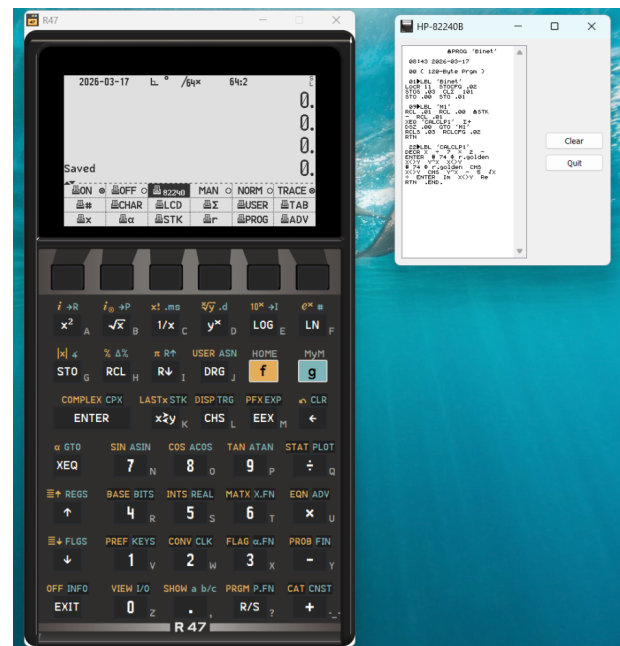


Figure 2: WP 34s QtPrinterEmulator

[Windows] Installation of the Windows HP82240B Printer Simulator from Christoph Giesselink

- Download the Installer package from <https://hp.giesselink.com/hp82240b.htm>
- Unzip and execute the HP82240Bv119Setup.exe
- If you get Microsoft Defender SmartScreen complaining that the application is unrecognized then click on 'More info' and then on 'Run anyway'
- Authorize the application to modify the content of your computer, this will bring up the Setup Wizard
- Click 'Next' four times, then click 'Install' and 'Finish'
- This will open the HP-82240B Printer simulator main window
- Click on 'Enabled' under 'Graphic' to open the HP-82240B Graphic window. This will open a second window for the Printer simulator providing a printout corresponding to the actual printer output (the main window shows only the characters that are in the printer font, not the characters sent as graphics).
- Launch the C47 or R47 simulator
- In the C47/R47 simulator **PRINT** menu select  **ON** to enable printing
- Now all printing commands will be directed to the HP-82240B Printer simulator
- Adjust the Printer simulator Graphic window size to the width and length that suits your usage
- A right click in the Graphic window will bring a menu with options to :
 - 'Copy' the content of the window
 - 'Advance' the paper
 - 'Clear' the window

[Linux] Installation of the WP 34s Qt Printer Emulator

- Download the Linux 64-bit package (wp-34s-emulator-linux64.tgz) from <https://sourceforge.net/projects/wp34s/files/emulator/>
- Uncompress wp-34s-emulator-linux64.tgz
- Install the QT GUI library if QT is not installed on your machine. The instructions below have been tested on Zorin OS and should be valid also for Ubuntu and it's derivatives such as Linux Mint:
 - `sudo add-apt-repository ppa:rock-core/qt4`
 - `sudo apt-get update`
 - `sudo apt-get install libqtgui4`
- Run HP-82240B to launch the printer emulator
- In the C47/R47 simulator **PRINT** menu select  **ON** to enable printing
- Adjust the Printer simulator Graphic window size to the width and length that suits your usage , increasing the width will increase automatically the character size once the window is large enough to accommodate the next zoom factor.
- Now all printing commands will be directed to the HP-82240B Printer simulator

Appendix D: Printers used to develop & test the C47/R47 printing functions



Figure 3: From left to right: HP-82240B, Martel MCP8850B, Martel MCP7850B and Midtronics 182-003B (a custom Martel MCP1830)

	HP-82240B	Martel MCP8850B	Martel MCP7850B	Midtronics 182-003B
Printing system	Thermal moving head	Thermal line head system	Thermal line head system	Thermal line head system
Texte line	8 x 166 dots	24 x 384 dots	24 x 384 dots	24 x 384 dots
Characters per line	24	48 max (24 default)	48 max (24 default)	48 max (24 default)
Buffer size	200 characters	6 Kbytes	5 Kbytes	5 Kbytes
Paper width	57 mm	58 mm	58 mm	58 mm
Character set	Roman- 8, ECMA-94	Roman 8, ECMA 94	Roman 8, ECMA 94	Roman 8, ECMA 94
Interface	HPIR	HPIR	HPIR	HPIR, IrDA, RS-232
Power	4 x 1.5V alkaline, AA cells	4 x 1.5V alkaline, AA cells	4 x 1.5V alkaline, AA cells	4 x AA NiMh batteries
Dimensions	90 x 185 x 61 mm	91 x 185 x 58 mm	85.5 x 150 x 55 mm	86 x 148.8 x 57.5 mm
Weight (with paper and batteries)	463 g	425 g	400 g	395 g

Appendix E: References

- [Hewlett-Packard Journal October 1987](#)
- The HP 82240 thermal printer series: <https://www.keesvandersanden.nl/calculators/hp82240.php>
- HP 82240A Infrared Printer Operating Instructions: <https://literature.hpcalc.org/items/488>
- HP 82240B Infrared Printer Operating Instructions: <https://literature.hpcalc.org/items/489>
- Martel [MCP8850B HPIR THERMAL PRINTER User Guide](#)
- Martel [MCP7850 Infra-Red THERMAL PRINTER Series Applications Datasheet](#)
- Midtronics [182-003A IR/RS232 THERMAL PRINTER User Guide](#)
- Midtronics [182-003B IR/RS232 THERMAL PRINTER User Guide](#)

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